## Arithmetic sequences

**1** Determine whether each of the following sequences is an arithmetic progression:

**a** $1, \sqrt{5}, 5, 5\sqrt{5}, \ldots$

**b** $4, 7, 10, 13, \ldots$

**c** $2, 2^2, 2^4, 2^6, \ldots$

**d** $2, 0, -2, -4, \ldots$

**e** $3, -3, 3, -3, \ldots$

**f** $3, 6, 12, 24, \ldots$

**g** $-32, -26, -20, -14, \ldots$

**h** $5, 7, 5, 7, \ldots$

**2** How is the common difference of an arithmetic sequence obtained?

**3** Find the common difference of the following arithmetic sequences:

**a** $2, 0, -2, -4, \ldots$

**b** $7, 18, 29, \ldots$

**c** $330, 280, 230, 180, \ldots$

**d** $-6, -\frac{39}{7}, -\frac{36}{7}, -\frac{33}{7}, \ldots$

**4** Write down the next two terms of the following arithmetic sequences:

**a** $4, 8, 12, \ldots$

**b** $6, 2, -2, -6, \ldots$

**c** $9, 14, 19, \ldots$

**d** $3.5, 5, 6.5, \ldots$

**5** Find the missing terms in the following arithmetic progressions:

**a** $8, \square, 16, 20, \square$

**b** $\square, 0, \square, 10, \square$

**6** For each of the following, write the first four terms in the arithmetic progression.

**a** The first term is $-10$ and the common difference is $4$.

**b** The first term is $-8$ and the common difference is $-2$.

**c** The first term is $a$ and the common difference is $d$.

**7** Consider the arithmetic sequence $1.4, 2.3, 3.2, \ldots, 10.4$

**a** Determine $d$, the common difference.

**b** Find $n$, the number of terms in the sequence.

**8** Insert five terms in the arithmetic sequence which has $-12$ as its first term and $24$ as its last term:

$$-12, \square, \square, \square, \square, \square, 24$$

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**9** Find the value of $x$ such that $x+4, 6x+5$, and $9x-8$ form successive terms in an arithmetic progression.

**10** An arithmetic progression has $u_5 = 21$ and $u_{19} = 77$.

- **a** Find the value of $d$.
- **b** Find the value of $u_1$.
- **c** Find $u_{11}$, the $11$th term in the sequence.

**11** An arithmetic progression has $u_6 = -11$ and $u_8 = -17$.

- **a** Find the value of $d$.
- **b** Find the value of $u_1$.
- **c** Hence find the $14$th term in the progression.

**12** In an arithmetic progression, the first term is $32$. Given that the $9$th term is $4$ times the $5$th term, find the common difference $d$.

**13** The first three terms of an arithmetic progression are: $34, 27, 20, \ldots$

- **a** Find the range of values of $n$ for which the terms in the progression are positive.
- **b** Hence determine the number of positive terms in the progression.
- **c** Find the last positive term in the progression.

**14** In an arithmetic progression the $6$th term is $x$ and the $10$th term is $y$.

- **a** Form an expression for $d$ in terms of $x$ and $y$.
- **b** Form an expression for $a$ in terms of $x$ and $y$.
- **c** Hence, find an expression for the $16$th term in terms of $x$ and $y$.

**15** The $14$th term of an arithmetic progression is equal to the sum of the $5$th term and the $10$th term. The common difference is $-2$.

- **a** Find the value of $u_1$.
- **b** Hence, find the $9$th term.

**16** Consider the sequence $2, 7, 12, \ldots$. If the $n$th term is $132$, find the value of $n$.

**17** Consider the sequence $2, -3, -8, \ldots$. If the $n$th term is $-578$, find the value of $n$.

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## Explicit rules

**18** For each of the following sequences:

i Determine $d$, the common difference.

ii State the expression for finding $u_n$, the $n$th term in the sequence.

iii Determine $u_{10}$, the 10th term in the sequence.

**a** $5, 12, 19, \ldots$ **b** $17, 16.2, 15.4, \ldots$ **c** $10, 3, -4, \ldots$ **d** $5, \frac{23}{4}, \frac{13}{2}, \ldots$

**19** For each of the following definitions of the $n$th term of a sequence:

i List the first four terms of the sequence.

ii Find the common difference.

**a** $u_n = 3n + 8$ **b** $u_n = 11 + (n-1) \times 10$

**c** $u_n = -7 - 3(n-1)$

**20** For the each of the following formulas for the $n$th term of an arithmetic progression:

i Find $u_1$, the first term of the sequence.

ii Find $d$, the common difference.

iii Find $u_5$, the 5th term in the sequence.

**a** $u_n = 15 + 5(n-1)$ **b** $u_n = -8n + 28$

**c** $u_n = 25 - 4(n-1)$ **d** $u_n = 12n - 9$

**21** Consider an arithmetic progression where $u_7 = 44$ and $u_{14} = 86$.

**a** Determine $d$, the common difference.

**b** Determine $u_1$, the first term in the sequence.

**c** State the equation for $u_n$, the $n$th term in the sequence.

**d** Hence find $u_{25}$, the 25th term in the sequence.

## Applications

**22** A termite treatment will cost $\$250$ for the first half hour, $\$245$ for the second half hour, $\$240$ for the third half hour and so on. Find the cost of a treatment that takes 6 hours.

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**23** A ball starts rolling down a slope. It rolls $25$ cm during the first second, $53$ cm during the second second, $81$ cm during the third second and so on. At this rate, what is the total distance it will have rolled after $10$ seconds?

**24** A diving vessel descends below the surface of the water at a constant rate so that the depth of the vessel after $1$ minute, $2$ minutes and $3$ minutes is $50$ metres, $100$ metres and $150$ metres respectively.

**a** By how much is the depth increasing each minute?

**b** What will the depth of the vessel be after $4$ minutes?

**c** Continuing at this rate, what will be the depth of the vessel after $10$ minutes?

**25** When a new school first opened, $a$ students started at the school. Each year, the number of students increases by the same amount, $d$.
At the beginning of its $7$th year, it had $361$ students. At the end of the $11$th year, the school had $536$ students.

**a** Find $d$, the number of students who joined the school each year.

**b** Find $u_1$, the number of students that started at the school when it first opened.

**c** At the end of the $n$th year, the school has reached its capacity at $921$ students. Find the value of $n$.

**26** A car bought at the beginning of 2009 is worth $\$1500$ at the beginning of 2015. The value of the car has depreciated by a constant amount of $\$50$ each year since it was purchased.

**a** What was the car purchased for in 2009?

**b** Write an explicit rule for the value of the car after $n$ years.

**c** Find the year in which the car will be worth half the price it was bought for.

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